

PROJECT IDEA • BACKGROUND • SIGNIFICANCE • LITERATURE SURVEY

A course-scale Minimum Viable Prototype (MVP) for demonstrating autonomous local navigation without relying on GPS as the core navigation source.

Core MVP Pipeline

Onboard Sensors → Local State Estimation → Goal Navigation → Confidence/Safety Decision → Obstacle Response → PX4 Flight Control

Project positioning

The project deliberately avoids presenting itself as a research-grade autonomous aircraft. The MVP focuses on a small, understandable and demonstrable ROS 2 architecture that captures the essential navigation loop.

1. PROJECT IDEA

The proposed project is a ROS 2-based autonomous drone navigation MVP designed to demonstrate local navigation in a GPS-independent environment. Instead of using an external global positioning source as the primary navigation reference, the drone establishes a local coordinate frame relative to its starting point and estimates its movement using onboard sensing.

The MVP combines an IMU with a complementary local-motion measurement such as optical flow, together with a range sensor for altitude and/or obstacle-distance information. These measurements feed a local state-estimation layer. A goal-navigation node then converts a relative destination into simple movement commands.

A lightweight confidence-aware decision layer is proposed between state estimation and navigation. When measurements are reliable, the system operates normally. When confidence degrades, it can reduce motion aggressiveness. When confidence or safety becomes unacceptable, it can enter a safe-hold state.

MVP concept

Input	Processing	Decision	Output
IMU + optical flow/local motion + range	Local state estimation + confidence assessment	Goal navigation + obstacle safety	Move / slow / stop / safe-hold

Proposed operating principle

Navigate when confident. Slow when uncertain. Stop when unsafe.

2. BACKGROUND

Unmanned Aerial Vehicles increasingly operate in inspection, disaster response, exploration, monitoring and other environments where autonomous mobility can provide an advantage. Conventional waypoint navigation generally benefits from GNSS/GPS because it supplies a global position reference.

However, GNSS availability is not guaranteed. Research reviews identify indoor spaces and other constrained environments where GNSS can be unavailable or unreliable. GNSS can also be vulnerable to interference such as jamming and spoofing. The navigation problem therefore becomes more than simply obtaining a global coordinate: the vehicle must perceive its environment, estimate its state, plan motion and control the vehicle.

Removing GPS from the navigation loop creates a state-estimation challenge. IMUs provide high-rate acceleration and angular-rate measurements, but dead reckoning accumulates error. Research therefore combines inertial measurements with complementary sources such as vision, optical flow, LiDAR, UWB or altitude measurements.

For this course MVP, the goal is not to reproduce a complete SLAM system. Instead, the project isolates the essential pipeline: obtain local motion information, estimate a usable local state, move toward a relative goal, detect a nearby obstacle and respond safely.

3. SIGNIFICANCE

Why is it required?

A navigation system that does not depend on external GPS infrastructure can remain useful where satellite positioning is unavailable or unreliable. More importantly for a ROS course, it demonstrates that autonomous navigation requires sensing, state estimation, decision making, navigation and safety behavior working together.

How is it useful?

Search & Rescue	Indoor or partially collapsed structures where GNSS reception is unavailable.
Industrial Inspection	Warehouses, plants, tunnels and other constrained environments.
Navigation Resilience	Local navigation remains useful when external positioning becomes unreliable.
Robotics Education	Demonstrates ROS 2 sensing, estimation, communication, navigation and safety in one integrated MVP.

4. LITERATURE SURVEY

The selected research works show a progression from individual localization techniques toward complete navigation pipelines. They also reveal a trade-off between accuracy, computational complexity, sensing requirements and implementation scope.

Gyagenda et al. (2022)	GNSS-independent UAV navigation review
Method / finding	Reviews perception, localization and motion planning and emphasizes that navigation is a combination of components rather than localization alone.
Project relevance	Supports our separation of sensing, local state estimation and navigation.
DOI	10.1016/j.robot.2022.104069

Chang et al. (2023)	UAV autonomous navigation in GPS-denied environments
Method / finding	Provides a taxonomy of mapless and map-based navigation strategies and discusses measurable navigation performance.
Project relevance	Supports treating navigation as an end-to-end system with measurable behavior.
DOI	10.1016/j.robot.2023.104533

Youn et al. (2021)	Collision-free navigation using low-cost sensors
Method / finding	Uses RGB-D, 2D LiDAR, altimeter, ESKF, SLAM and collision-free path planning for autonomous GPS-denied flight.
Project relevance	Shows that low-cost sensing can support complete navigation, while illustrating the complexity of a research-grade system.
DOI	10.1007/s12555-019-0797-7

Qin, Li & Shen (2018)	VINS-Mono visual-inertial state estimation
Method / finding	Fuses pre-integrated IMU measurements with monocular image features for metric visual-inertial odometry.
Project relevance	Motivates complementary sensing, but full VIO is beyond the MVP target.
DOI	10.1109/TRO.2018.2853729

Campos et al. (2021)	ORB-SLAM3 visual / visual-inertial / multimap SLAM
Method / finding	Combines visual-inertial estimation, mapping and relocalization to improve robustness and long-term consistency.
Project relevance	Demonstrates the power of SLAM while justifying why full mapping is outside the course MVP.

DOI	10.1109/TRO.2021.3075644
Zhang & Singh (2014)	LOAM LiDAR odometry and mapping
Method / finding	Separates high-frequency odometry from lower-frequency mapping and registration for real-time LiDAR navigation.
Project relevance	Shows LiDAR's strength for geometric motion estimation, but full mapping is unnecessary for this MVP.
DOI	10.15607/RSS.2014.X.007

5. LITERATURE SYNTHESIS & PROJECT POSITIONING

Approach	Strength	Limitation for MVP
IMU / dead reckoning	Simple, high-rate, low-cost	Position drift accumulates over time
Optical flow + range	Lightweight local-motion aid	Depends on suitable measurement conditions
VIO	Metric visual-inertial estimation	Camera processing and calibration increase complexity
Visual SLAM	Mapping + localization + loop closure	Higher compute and software complexity
LiDAR odometry / SLAM	Strong geometry, less lighting dependence	Sensor and mapping burden is larger
UWB-aided navigation	External position constraints	Requires infrastructure

What our MVP takes from the literature

- **Navigation decomposition:** separate perception, localization and planning rather than treating navigation as one algorithm.
- **Drift management:** use complementary local measurements instead of relying on IMU-only dead reckoning.
- **Scope discipline:** avoid full SLAM/VIO/LiDAR mapping when the course objective is to demonstrate a working ROS 2 navigation pipeline.
- **Safety:** make obstacle response and low-confidence behavior explicit parts of the navigation architecture.
- **Engineering differentiation:** a compact confidence-aware ROS 2 integration, not a claim of a new localization algorithm.

Recommended project positioning

“Our project investigates a lightweight ROS 2-based GPS-independent navigation MVP that uses onboard sensing to maintain a local coordinate frame, navigate toward a relative goal and respond safely to obstacles. The contribution is practical integration and confidence-aware safety behavior, not a new localization algorithm.”

6. REFERENCES

- [1] Gyagenda, N., Hatilima, J. V., Roth, H., & Zhmud, V. (2022). A review of GNSS-independent UAV navigation techniques. *Robotics and Autonomous Systems*, 152, 104069. DOI: 10.1016/j.robot.2022.104069.
- [2] Chang, Y., Cheng, Y., Manzoor, U., & Murray, J. (2023). A review of UAV autonomous navigation in GPS-denied environments. *Robotics and Autonomous Systems*, 170, 104533. DOI: 10.1016/j.robot.2023.104533.
- [3] Youn, W., Ko, H., Choi, H., Choi, I., Baek, J. H., & Myung, H. (2021). Collision-free Autonomous Navigation of a Small UAV Using Low-cost Sensors in GPS-denied Environments. *International Journal of Control, Automation and Systems*, 19(2), 953–968. DOI: 10.1007/s12555-019-0797-7.
- [4] Qin, T., Li, P., & Shen, S. (2018). VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator. *IEEE Transactions on Robotics*, 34(4), 1004–1020. DOI: 10.1109/TRO.2018.2853729.
- [5] Campos, C., Elvira, R., Gómez Rodríguez, J. J., Montiel, J. M. M., & Tardós, J. D. (2021). ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM. *IEEE Transactions on Robotics*, 37(6), 1874–1890. DOI: 10.1109/TRO.2021.3075644.
- [6] Zhang, J., & Singh, S. (2014). LOAM: Lidar Odometry and Mapping in Real-time. *Robotics: Science and Systems X*. DOI: 10.15607/RSS.2014.X.007.

Source alignment

The uploaded sample frames the project around a self-generated relative coordinate system, IMU-based dead reckoning, optical-flow/local-motion sensing and lightweight GPS-independent navigation. This document preserves that framing while grounding the literature survey in verified research works and narrowing the proposed system to an achievable ROS-course MVP.